

Written Homework 3

Problem Solving, Math 1000 – Summer 2016

Due Friday, June 17th

Problem 1 Suppose that you deposit \$4000 in a savings account that pays 2% annual interest, with interest credited to the account at the end of each year. Assuming that no withdrawals are made and that interest is compounded annually, find the balance in the account after 3 years. Then, find the balance in the account after 5 years and 7 months.

The amount of money in the account after 33 years is \$4244.83. The amount of money in the account after 5 years and 7 months is \$4416.32.

Problem 2 For an investment of $P = \$1262.50$ compounded annually with a 6.75% APR, find the value of the investment after 2 years. Then, find the value of the investment after 3 and a half years.

The value of the investment after 2 years is \$1438.69. The value of the investment after 3 and a half years is \$1535.80.

Problem 3 Between the years 1994 and 2004, the average annual inflation rate was 2.7%. Find the salary in the year 2004 that would be equivalent to a \$35,000 salary in 1994.

The salary in 2004 dollars that would be equivalent to a \$35,000 salary in 1994 is \$45,684.88.

Problem 4 For 5 years Jane deposits \$200 per month into an account that pays 3% APR compounded monthly. Assuming Jane did not make any withdrawals and no bank fees were charged, how much interest does she have 8 years after her initial deposit?

(HINT: Split up your calculations into two parts – one for the first 5 years, then one for the last 3 years. What formula should you use for each?)

After 8 years, the interest is $14,145.36 - (200)(60) = \$2,145.36$

Problem 5 Congratulations! You just a baby named George! [If you were not expecting this or thought the baby might be named something other than George, also “Surprise!”] You have decided that George needs to go to college, but must have \$40,000 in a savings account before he starts.

- (a) How much money do you need to put in a savings account that pays 3% compounded quarterly (that is, 4 times per year) so that your George can go to college on his 18th birthday?

$$F = P(1 + p)^T$$

$$F = 40000$$

$$P = ?$$

$$p = \frac{0.03}{4} = 0.0075$$

$$T = 18 \times 4 = 72$$

$$\$23,356.95$$

- (b) If you put \$25,000 in a savings account that pays 2% compounded quarterly, how old is George before he can start college? $F = P(1 + p)^T$

$$F = 40000$$

$$P = 25000$$

$$p = \frac{0.02}{4} = 0.005$$

$$T = 4t$$

$$T = 94.24, t = 23.56, \text{ but since this is compounded quarterly need to go to the next quarter: } 23.75 \text{ years old}$$

- (c) If you put \$25,000 in a savings account that pays interest compounded monthly, what interest rate is needed so your George can go to college on his 16th birthday? $F = P(1 + p)^T$

$$F = 40000$$

$$P = 25000$$

$$p = \frac{r}{12}$$

$$T = 16 \times 12 = 192$$

$$p = 0.00245, 2.94\% \text{ APR}$$

Problem 6 Write as a decimal:

1. $0.35\% = 0.0035$

2. $350\% = 3.50$

Write as a percentage:

1. $0.35 = 35\%$

2. $350 = 35,000\%$

Problem 7 Oh, the difference a dollar makes! Suppose you run up a balance of \$10,000 on a credit card, with an interest rate of 22% compounded monthly.

(a) If you only pay \$184.33 per month, how long will it take to pay off the balance?

$$M = P \frac{p(1+p)^T}{(1+p)^T - 1}$$

$$M = \$184.33$$

$$P = \$10,000.00$$

$$p = \frac{0.22}{12}$$

$$T = ?$$

$$24 \text{ years } (T = 288 \text{ months})$$

(b) How much interest do you pay over this time span?

$$288 \text{ months} \times \$184.33 \text{ per month} - \$10,000 = \$43,087.04$$

(c) If you only pay \$183.34 per month, how long will it take to pay off the balance?

$$M = P \frac{p(1+p)^T}{(1+p)^T - 1}$$

$$M = \$183.34$$

$$P = \$10,000.00$$

$$p = \frac{0.22}{12}$$

$$T = ?$$

$$46 \text{ years, 11 months } (T = 563 \text{ months})$$

(d) How much interest do you pay over this time span?

$$563 \text{ months} \times \$183.34 \text{ per month} - \$10,000 = \$93,220.42$$

(e) If you only pay \$183.33 per month, how long will it take to pay off the balance? Will the balance ever be paid off?

$$M = P \frac{p(1+p)^T}{(1+p)^T - 1}$$

$$M = \$183.33$$

$$P = \$10,000.00$$

$$p = \frac{0.22}{12}$$

$$T = ?$$

At this point the payment is too small to keep up with the interest.

The balance will never be paid off!

Problem 8 *Oh, the difference a percent makes! Suppose you run up a balance of \$10,000 on a credit card, with an interest rate of 21% compounded monthly.*

(a) *If you only pay \$184.33 per month, how long will it take to pay off the balance?*

$$M = P \frac{p(1+p)^T}{(1+p)^T - 1}$$

$$M = \$184.33$$

$$P = \$10,000.00$$

$$p = \frac{0.21}{12}$$

$$T = ?$$

$$14 \text{ years } 4 \text{ months } (T = 172)$$

(b) *How much interest do you pay over this time span?*

$$172 \text{ months} \times \$184.33 \text{ per month} - \$10,000 = \$21,704.76$$

(c) *If you only pay \$183.34 per month, how long will it take to pay off the balance?*

$$M = P \frac{p(1+p)^T}{(1+p)^T - 1}$$

$$M = \$183.34$$

$$P = \$10,000.00$$

$$p = \frac{0.21}{12}$$

$$T = ?$$

$$14 \text{ years, } 11 \text{ months } (T = 179)$$

(d) *How much interest do you pay over this time span?*

$$179 \text{ months} \times \$183.34 \text{ per month} - \$10,000 = \$22,817.86$$

(e) *If you only pay \$183.33 per month, how long will it take to pay off the balance?*

$$M = P \frac{p(1+p)^T}{(1+p)^T - 1}$$

$$M = \$183.33$$

$$P = \$10,000.00$$

$$p = \frac{0.21}{12}$$

$$T = ?$$

$$14 \text{ years, } 11 \text{ months } (T = 179)$$